

Stan Serebriakov

Senior C++ Engineer

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cfr.pw, [github](https://github.com)

Experience

2024–Present **Senior C++ Engineer**, *MPK Soft*, St. Petersburg.

- Building a C++ imaging and rendering application: loupe and background estimation; a scripting layer embedding Qt's JavaScript engine to expose view control and the image-processing pipeline.
- Engineered a real-time object detection and tracking pipeline on embedded ARM: Hessian blob detection, MOSSE tracking, and automatic exposure and gain control.
- Built a wide-band signal-processing system: LoRa chirp recognition and frequency-hopping table estimation across a 500–1020 MHz span, with a data-processing server.

Stack: C++20, Qt, Boost, FlatBuffers, FFTW, OpenCV.

2021–2023 **Lead iOS Engineer**, *Constanta*, Moscow.

- Built a table-tennis augmented reality app with real-time ball tracking on Core ML, shipped from an Objective-C prototype to a published Swift application.
- Implemented core features, refactored legacy code for a leading sports analytics app.

Stack: Swift, C++, Core ML, Objective-C.

2015–2021 **Lead iOS Engineer**, *Luxoft*, St. Petersburg.

Delivered iOS applications for enterprise clients including Citibank, Gazprom Neft, and Gosuslugi; conducted hundreds of technical interviews worldwide.

Stack: Swift, Objective-C, OpenGL, MVVM, Lottie.

2015–2015 **Senior iOS Engineer**, *GuessMe*, St. Petersburg.

Built [burningbar](#), a Haskell code generator emitting Swift networking and JSON-mapping layers from an interface description, alongside feature work on a social networking app.

Stack: Haskell, Swift, Objective-C, SQLite.

2014–2014 **Lead R&D Engineer**, *x-Turion*, St. Petersburg.

Led software development for a compact home robot equipped with a surveillance camera for remote audio and video monitoring of private premises. Key device features included autonomous navigation, object and face recognition, and mobile app control.

Stack: C++, ROS, OpenCV.

2012–2014 **Mobile Engineer**, *i-Free*, St. Petersburg.

Built a marker-based augmented reality application on Qualcomm Vuforia, overlaying interactive content on banknotes, with a shared C++ core and OpenGL ES renderer targeting both iOS and Android.

Stack: C++, OpenGL ES, Objective-C, Java.

- 2009–2011 **Lead Engineer, Co-founder, Cybrox**, St. Petersburg.
Led the development of 3D IR-based head trackers and conducted research on webcam-based 6D head tracking.
- Implemented an OpenCV-based prototype and ported it to C for execution on a Blackfin DSP.
 - Co-authored the business plan and patent applications.
- Stack: C, OpenCV, OpenGL.
- 2009–2010 **Research Assistant, Jacobs University**, Bremen.
Implemented a 3D tracking system based on a time-of-flight camera: calibration, motion estimation, and a camera Ethernet driver. Co-authored A. Nüchter, S. Serebriakov, L. Stankevich: “Visual SLAM with a Time-of-Flight Camera”, *Journal of Optical Technology*, 2010.

Selected Projects

- 2025–Present **aktis** — physically based path tracer, github.com/cfr/aktis.
Monte Carlo renderer in C++20.
- Whitted, analytic direct, and path-tracing integrators with next event estimation, multiple importance sampling, and Russian roulette.
 - Modified Phong and GGX BRDFs, dielectrics with Fresnel and delta-lobe sampling, and a general quadric primitive (sphere, cylinder, cone, paraboloid).
 - Binned SAH bounding volume hierarchy; 10–30% speedup from a profiling-driven refactor.
 - Containerised CI running clang-tidy/clang-format and render-regression tests diffing output against base.

Education

- 2010–2012 **PhD Student in Applied Math, St. Petersburg Polytechnic University**.
Monocular 3D Tracking
- 2004–2010 **MSc in Robotics, St. Petersburg Polytechnic University**.
Simultaneous Localization and Mapping with a Time-of-Flight Camera

Certificates

- CSE167x: Computer Graphics, UC San Diego (edX), 2026
- CSE 168: Computer Graphics II – Rendering, UC San Diego (edX), 2026

Skills

- Languages C++20, C, GLSL, Swift, Objective-C, Haskell. Familiar with Go, Java, Python
- Graphics Physically based rendering, Monte Carlo path tracing, BVH acceleration, OpenGL/ES
- Other Linux, CMake, Qt, Computer Vision, DSP, $\text{\LaTeX}$